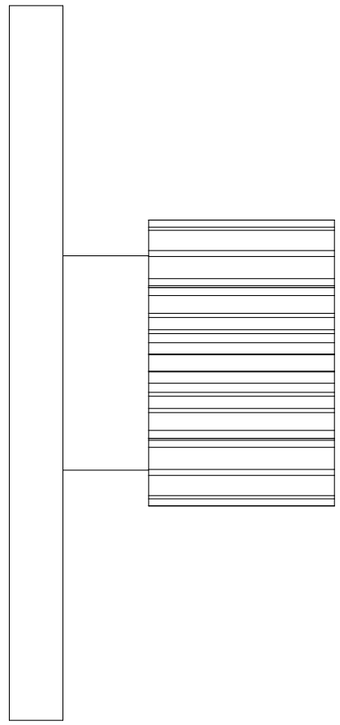
VAYN shoulder · milling drawing · 18 Sun-1

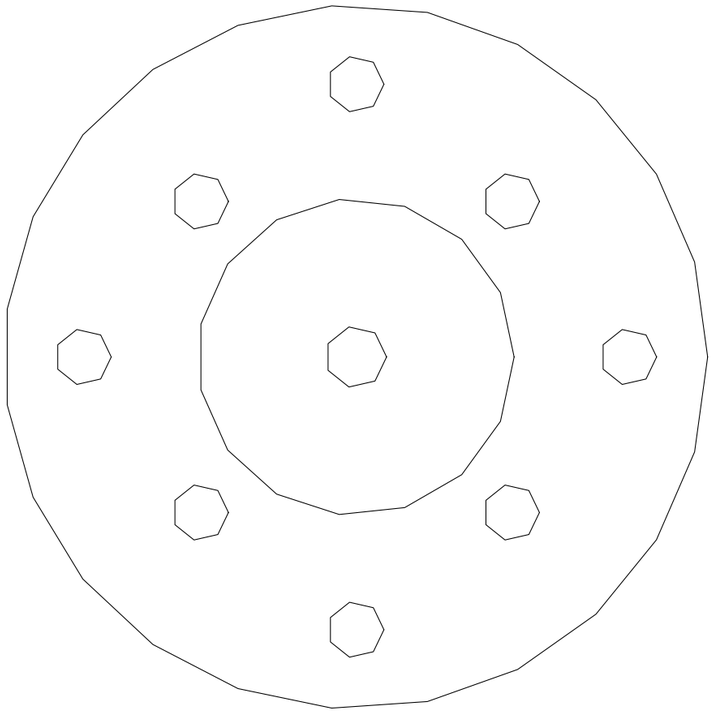
Pack A · geometry Rev 1.4 · qty 1 actuator

FRONT · Z horizontal, +X up



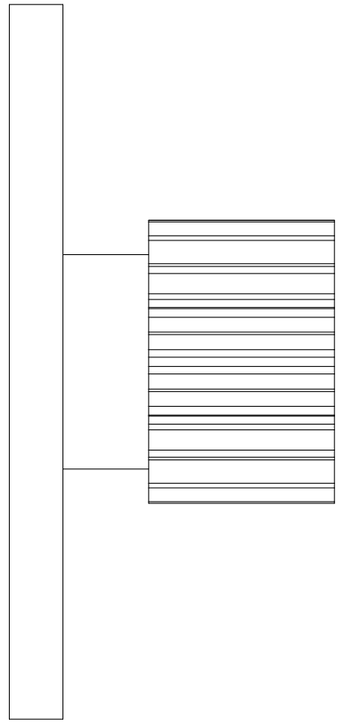
scale 2.91:1 view span 18.2 × 40.0 mm

END · from the arm side (+Z)



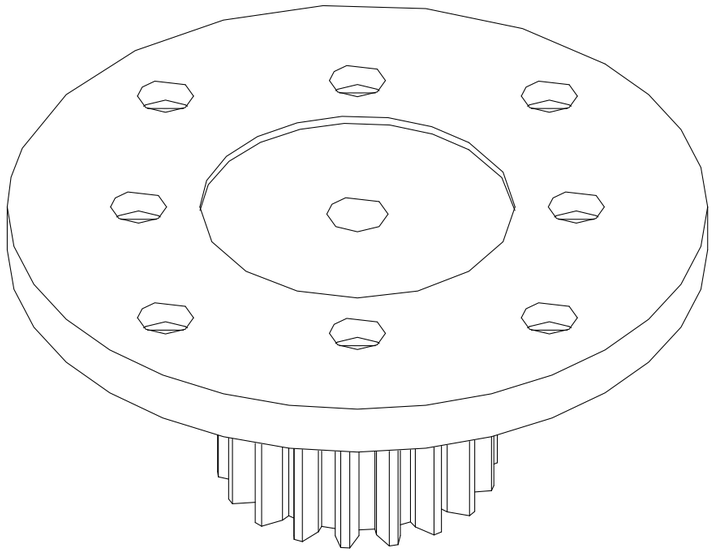
scale 2.87:1 view span 39.8 × 39.9 mm

SIDE · Z horizontal, +Y up



scale 2.91:1 view span 18.2 × 40.0 mm

ISOMETRIC



scale 2.85:1 view span 40.0 × 31.0 mm

18

Sun-1

Quantity in this actuator: 1
Material: 16MnCr5
Group: Gears
Finished size: Flange Ø40 × 3 mm, teeth Ø16 × 10.4 mm, overall 18.2 mm
Mass: 42.38 g each
Model bounding box, X × Y × Z: 40.0 × 40.0 × 18.2 mm
Raw stock: Ø45 × 30 mm 16MnCr5, annealed, same heat as the other 16MnCr5
Process: Turn, hob or shape
Heat treatment / finish: Carburize, case 0.12–0.18 mm, 58–62 HRC, core 30–40 HRC. Hard-finish the teeth.
STEP solid: 18 Sun-1 (16MnCr5)

Controlled features

- 18 teeth, module 0.8, 20° pressure angle, profile shift 0.
- Span over 3 teeth W3 = 6.082 mm, minus 0.015 / plus 0.
- ISO 1328 grade 6–7. Backlash 0.05 ± 0.02 mm per mesh.
- 4 × M3 countersunk on Ø31 mm, and 4 × M3 countersunk on Ø25 mm at 45°. Use the pattern the rotor has.
- Bore Ø3.5 × 10.5 mm for the motor stub. Hub relief Ø18 × 0.5 mm.
- Flange Ø40 h8.

STEP file is the geometric master. Do not scale this sheet. Sizes not listed here follow ISO 2768-1 class m and ISO 2768-2 class K. Drawing 18_sun_1.pdf